
```
clc;
clear;
close all;
```

Parameters

```
N = 100;           % Number of bits
sps = 20;          % Samples per symbol
SNR = 10;          % Signal-to-Noise Ratio (dB)
```

Generate Random Binary Data

```
data = randi([0 1], 1, N);
```

NRZ Encoding

0 -> 0 1 -> 1

```
tx_signal = repelem(data, sps);
```

Pass Signal Through AWGN Channel

```
rx_signal = awgn(tx_signal, SNR, 'measured');
```

Time Axis

```
t = (0:length(tx_signal)-1) / sps;
```

Plot Results

```
figure('Name', 'NRZ Encoding and AWGN', 'NumberTitle', 'off');
```

```
% Original Binary Data
```

```
subplot(4,1,1);
stairs(data, 'LineWidth', 1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);
```

```
% Transmitted NRZ Signal
```

```
subplot(4,1,2);
plot(t, tx_signal, 'LineWidth', 1.5);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

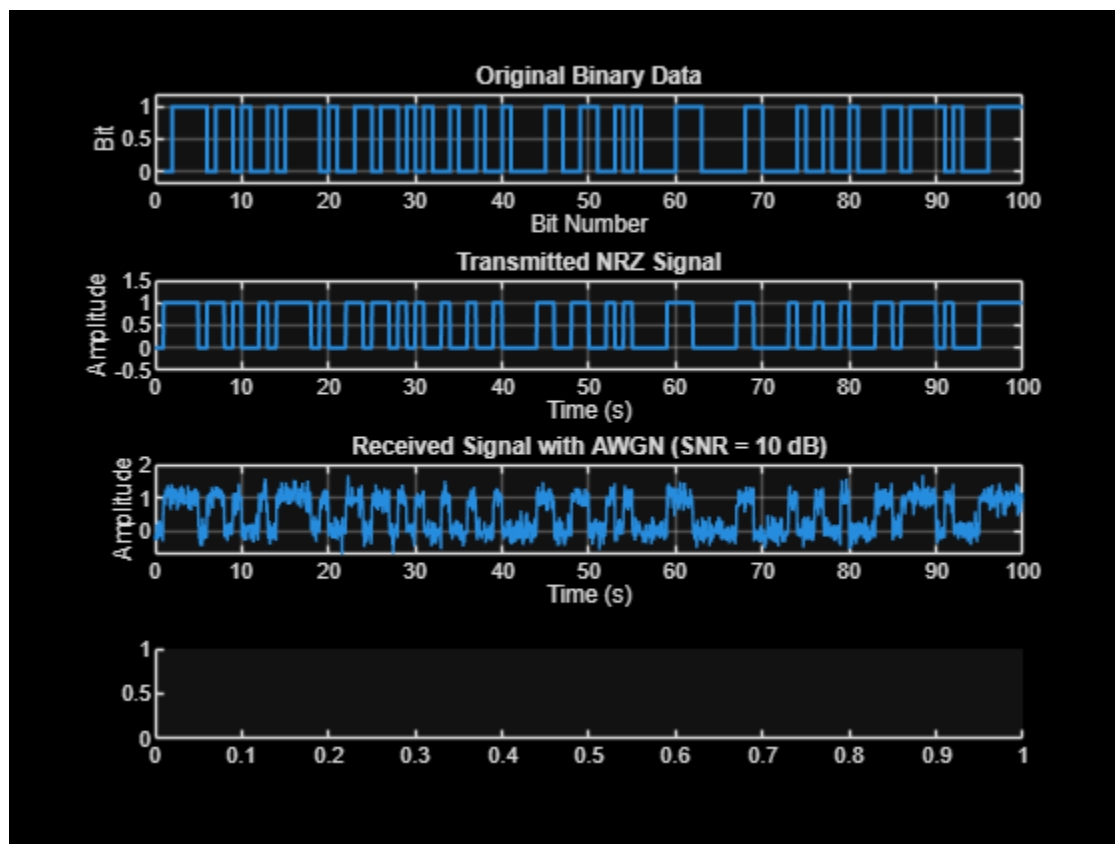
```

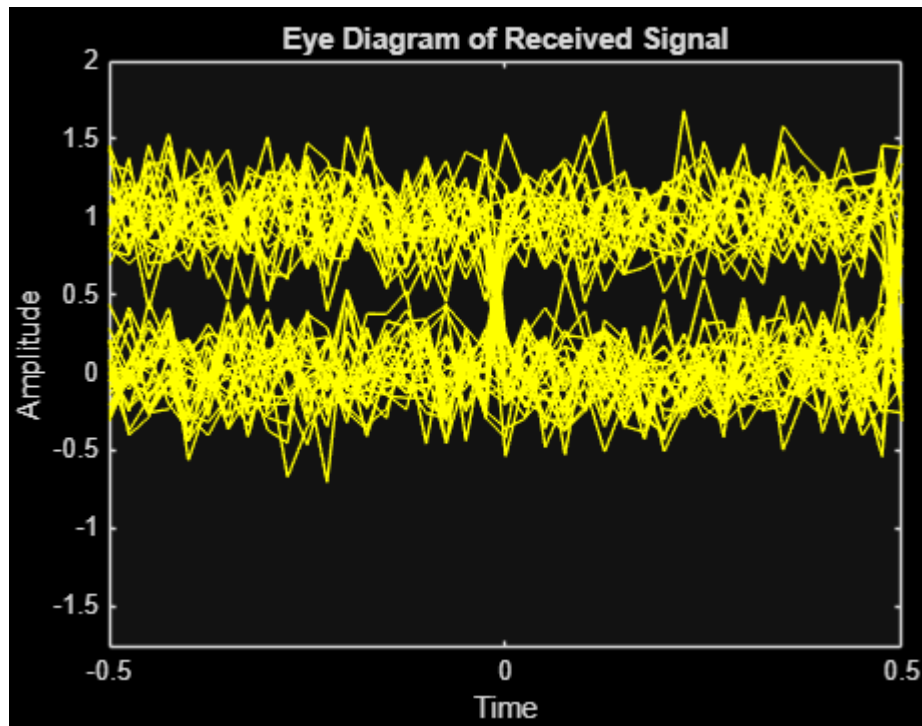
ylim([-0.5 1.5]);

% Received Signal
subplot(4,1,3);
plot(t, rx_signal);
grid on;
title(['Received Signal with AWGN (SNR = ', ...
      num2str(SNR), ' dB)']);
xlabel('Time (s)');
ylabel('Amplitude');

% Eye Diagram
subplot(4,1,4);
eyediagram(rx_signal, 2*sps);
title('Eye Diagram of Received Signal');

```





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